DeepHotspot-Seq: Sequence-Only Prediction of PPI Hotspots

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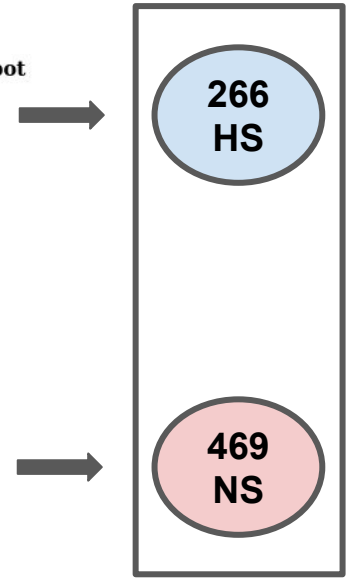
Dataset

PPI-HotSpot^{DB} (2021)

Uniprot ID	Uniprot Res ID	PDB ID	Chain	Resolution	PDB Res ID	Residue type	Mutations	Conservation	Pubmed	DDG	Hotspot
P21836	328	1c2b	A	4.5	297	F	I	1	8157652	1.98	P
P21836	155	1c2b	A	4.5	124	Y	Q	2	8157652	2.88	P
P21836	103	1c2b	A	4.5	72	Y	N	1	8157652	5.01	P
P21836	317	1c2b	A	4.5	286	W	R	1	8157652	8.45	P

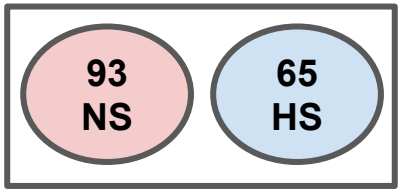
SKEMPI V2 (2019)

PDB Code	Mutation Pdb	Mutation Cleaned	Location	Hold Out Type	Hold Out Proteins	Affinity Mut	Affinity Wt	Ref.	Protein 1	Protein 2	T (°K)
1REW_AB_C	AA34D,DA53A,AB34D...	AA23D,DA42A,AB23D...	SUR,COR,SUR,SUR		1REW_AB_C,2QJ9_AB...	4.89E-10	6.67E-10		Bone morphogenetic protein-2	BMPR-IA receptor	298
1N8Z_AB_C	YB105F	YB105F	COR	AB/AG	AB/AG	1.4E-09	3.5E-10		Herceptin	erbB-2	310



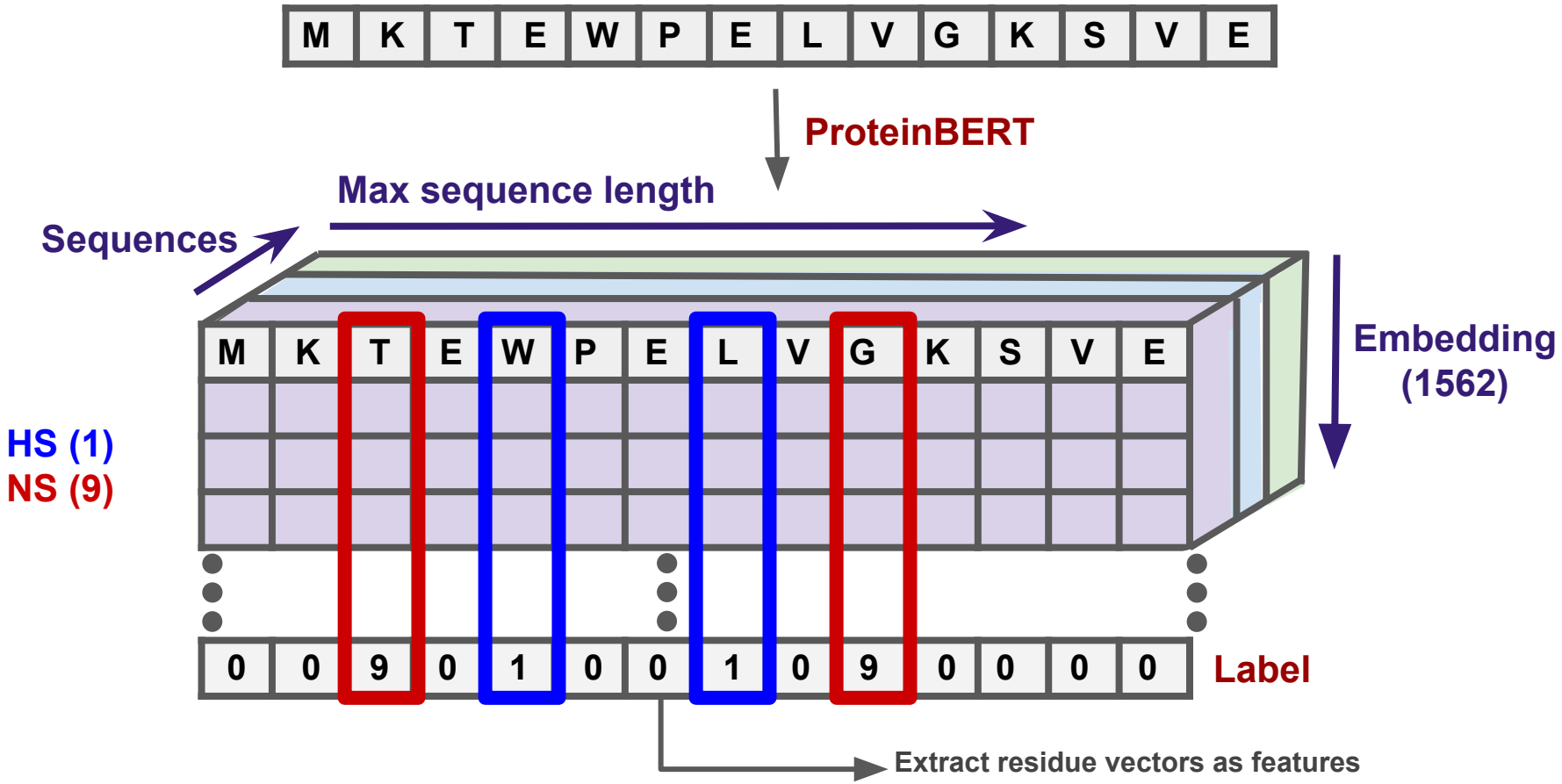
Training Set

BID (2018)



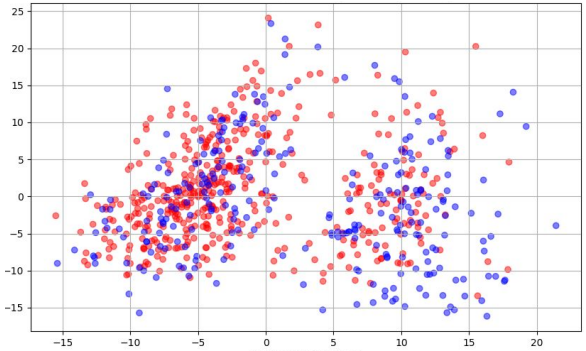
Independent Test Set

Feature Vector

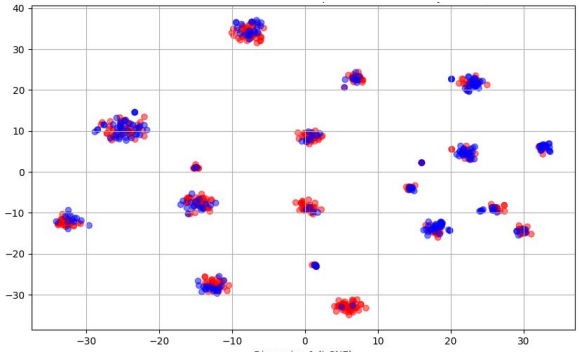


Embedding Clustering

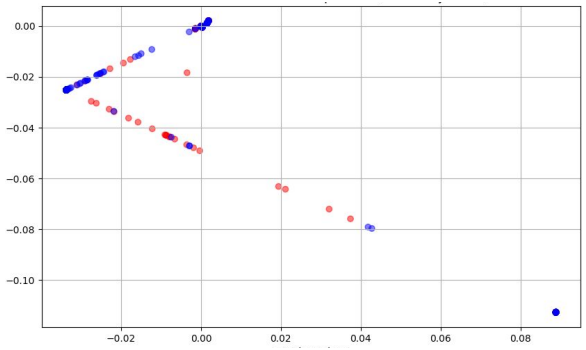
PCA



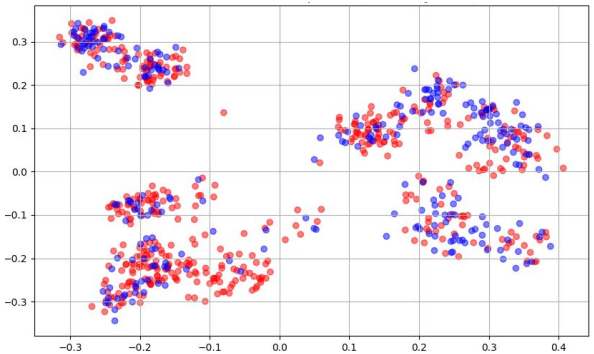
tSNE



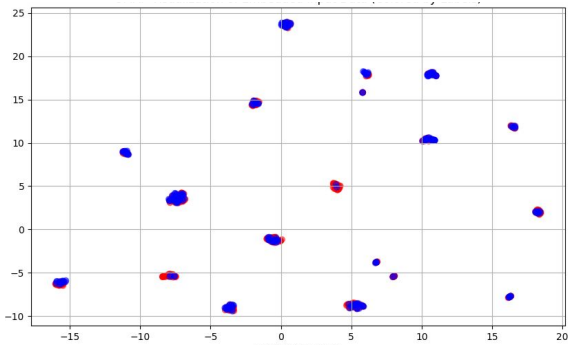
LLE



kernel-PCA



UMAP



Hotspot Nullspot

PCA: Principal Component Analysis

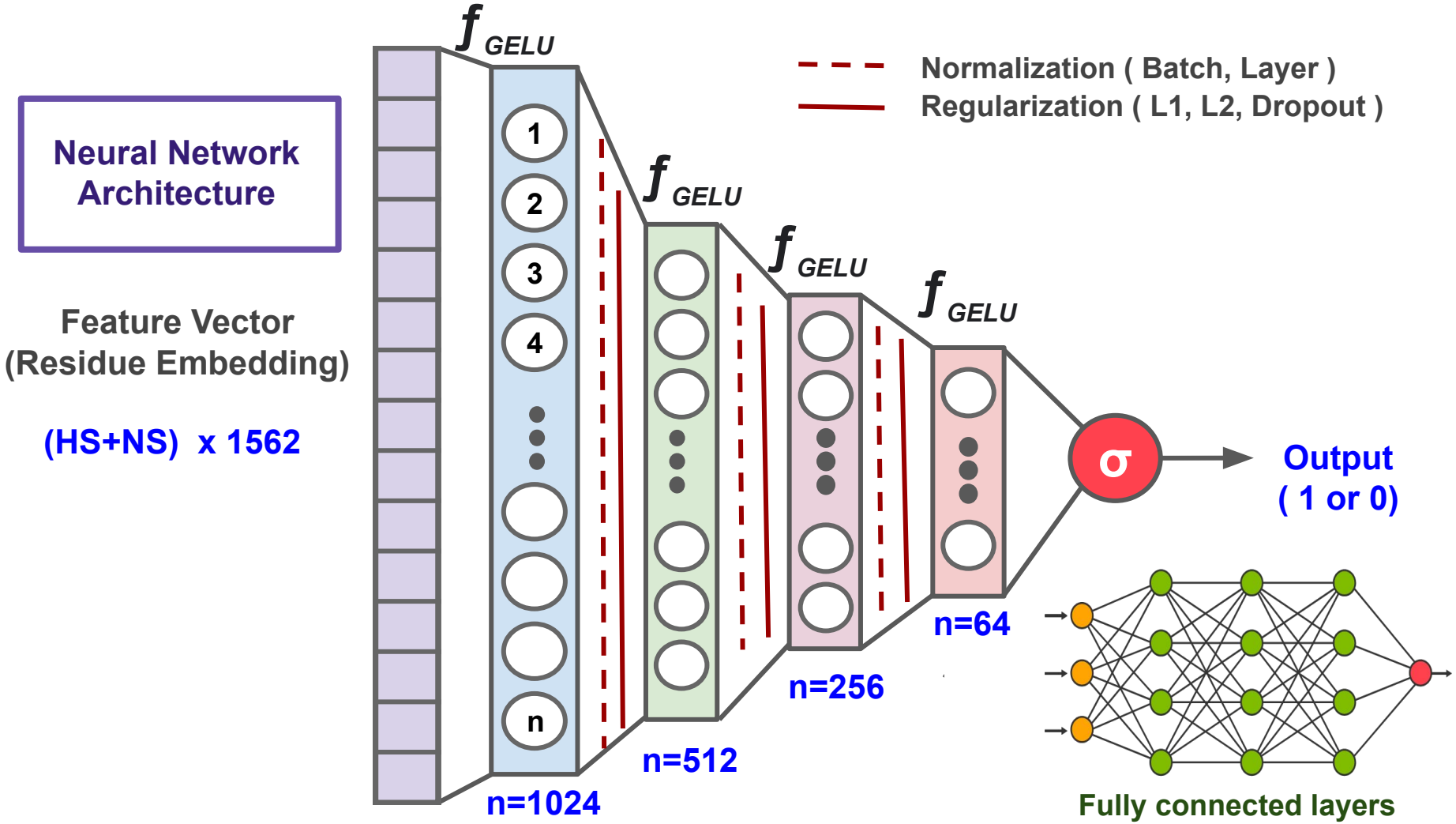
t-SNE: t-distributed Stochastic Neighbor Embedding

LLE: Locally Linear Embedding

UMAP: Uniform Manifold Approximation and Projection

Testing Common ML Algorithms

Classifier	10 Fold Cross Validation		Performance on BID Dataset		
	Mean F1 Score	Mean MCC	F1 Score	MCC	Accuracy
Random Forest	0.66	0.58	0.30	0.18	0.63
Gradient Boosting	0.82	0.75	0.31	0.01	0.55
SVM	0.77	0.65	0.25	-0.05	0.53
K-Nearest Neighbors	0.50	0.34	0.46	0.25	0.65
Decision Trees	0.65	0.45	0.44	0.04	0.53
Naive Bayes	0.59	0.37	0.56	0.17	0.58
MLP Neural Network	0.81	0.7	0.47	0.15	0.60
AdaBoost	0.76	0.64	0.41	0.07	0.56
XGBoost	0.82	0.73	0.37	0.03	0.55


Neural Network Architecture
Feature Vector (Residue Embedding)
(HS+NS) x 1562
f GELU
f GELU
f GELU
f GELU
1
2
3
4
n
n=1024
n=512
n=256
n=64
σ
Output (1 or 0)
Normalization (Batch, Layer)
Regularization (L1, L2, Dropout)
Fully connected layers

Hyperparameter Tuning

Hyperparameter values to search over (Grid Search)

1. regularization_types = ['l1', 'l2', 'none']
2. dropout_rates = [0.0, 0.3, 0.5]
3. l1_l2_strengths = [0.01, 0.05, 0.1]
4. learning_rates=[0.1, 0.01, 0.001]

Total params : 2,286,529
Trainable params : 2,282,753
Non-trainable params : 3,776

reg_ type	strength	dropout _ rate	learning _rate	f1 score	precision	recall
l1	0.01	0	0.001	0.733	0.771	0.698
l2	0.01	0.3	0.001	0.722	0.709	0.736
l2	0.01	0	0.001	0.708	0.667	0.755
l2	0.05	0.3	0.001	0.706	0.735	0.679
l1	0.01	0.3	0.001	0.701	0.773	0.642
l1	0.05	0	0.001	0.696	0.645	0.755
none	0	0.5	0.001	0.678	0.603	0.774

The Challenge of Sequence-Only Prediction

- Directly predicting hotspots from sequences is difficult due to the **lack of explicit 3D structural context.**
- Objective: Leverage established structural tools to **create a structure-guided training set** for the Deep Learning model.

In-House Structure-based Hotspot Predictors

1. **Grahotspot**: Identifies hotspots using graph theoretical parameters to assess residue centrality at protein interfaces.
2. **HotPIES**: An in-house machine learning tool integrating 18 features, including non-covalent energies, solvent accessibility (ASA), and secondary structure.
3. **PPCheck**: Analyzes residues based on energy contribution and spatial proximity within the complex.

The Consensus Strategy
(Grahotspot ∩ HotPIES ∩ PredHS2)

- Implemented a strict intersection logic: A residue is labeled a **"Hotspot"** only **if predicted by all three structural tools.**
- This "Consensus" approach prioritizes label precision over quantity to minimize noise in the training data.
- Resulted in **759 high-confidence consensus hotspots** for model training

Data Augmentation via PIMADb

- Used the **PIMADb Dimers + Trimers dataset** to expand the structural diversity of the training set.
- Allows the sequence-only model to learn from labels derived from a wide variety of protein-protein interface geometries
- Labels generated from structural parameters are then used to train the sequence-only model using ProteinBERT embeddings

Data augmentation with consensus hotspots

PIMADb Dimers + Trimers Dataset

↓

Grahotspot ∩ HotPIES ∩ PredHS2

↓

Total consensus hotspots = 759

Test Performance :

F1 Score : 53.93 %

Precision : 66.67 %

Recall : 45.28 %

Accuracy : 72 %

Performance on BID dataset:

F1-score : 58.1 %

Precision : 43 %

Recall : 91 %

MCC : 0.086

Accuracy : 46 %